C3.1 Integration of body systems

NAME	DATE	TOTAL
		_____ / 55

About 55 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

10 MARKS

1 DEFINE emergent property. [C3.1.2] [1]

2 STATE the gland that secretes melatonin. [C3.1.11] [1]

3 STATE the gland that secretes epinephrine. [C3.1.12] [1]

4 DISTINGUISH between a neuron and a nerve. [C3.1.8] [2]

5 STATE two roles of the cerebellum. [C3.1.10] [2]

6 STATE what baroreceptors monitor, and one place where they are found. [C3.1.14] [2]

7 IDENTIFY the part of the nervous system that coordinates peristalsis in the small intestine. [C3.1.16] [1]

Section B: Understanding

21 MARKS

8 COMPARE AND CONTRAST nervous and hormonal signalling. [C3.1.3] [4]

NERVOUS	HORMONAL

9 **OUTLINE** how the brain acts as a central integrating organ. [C3.1.4] [3]

10 **OUTLINE** how a message passes from a touch receptor in the skin to the cerebral hemispheres, and then to a muscle. [C3.1.6, 7] [3]

11 **DRAW** a labelled diagram of a pain reflex arc, including the spinal cord. [C3.1.9] [5]

12 **EXPLAIN** how the effects of epinephrine prepare the body for intense muscle contraction. [C3.1.12] [3]

13 **OUTLINE** how the hypothalamus controls the endocrine system. [C3.1.13] [3]

Section C: Working with data

16 MARKS

All data in this section are invented for practice. The patterns are realistic.

14 [C3.1.14, C3.1.15]

A student's heart rate and ventilation rate were measured before, during and after 10 minutes of running.

TIME / MINUTES	0	5	10	12	15	20
HEART RATE / BEATS PER MINUTE	70	140	155	120	95	75
VENTILATION RATE / BREATHS PER MINUTE	14	28	34	26	18	15

a **CALCULATE** the percentage increase in heart rate from 0 to 10 minutes. [2]

b **DESCRIBE** the changes in ventilation rate during and after exercise. [2]

c **EXPLAIN** how the increase in ventilation rate is brought about. [4]

d **EXPLAIN** why heart rate falls after exercise ends. [3]

15 [C3.1.11]

The table shows typical blood melatonin concentration over one day, in a person sleeping from 23:00 to 07:00.

TIME	12:00	18:00	22:00	02:00	06:00	09:00
MELATONIN / PG CM^{-3}	5	8	30	70	40	8

a **STATE** the time at which melatonin concentration was highest. [1]

b **EXPLAIN** the pattern shown. [2]

c **SUGGEST** why flying across several time zones disturbs sleep for a few days. [2]

Section D: Longer answers

8 MARKS

16 **DESCRIBE** how heart rate is regulated in response to changes in blood pressure and blood chemistry. [6]

[C3.1.14]

17 **DISTINGUISH** between voluntary and involuntary control of movement of food through the gut. [2]

[C3.1.16]